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Depth-dependent surface wave dispersion equation calculated by the impedence matrix method

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Abstract

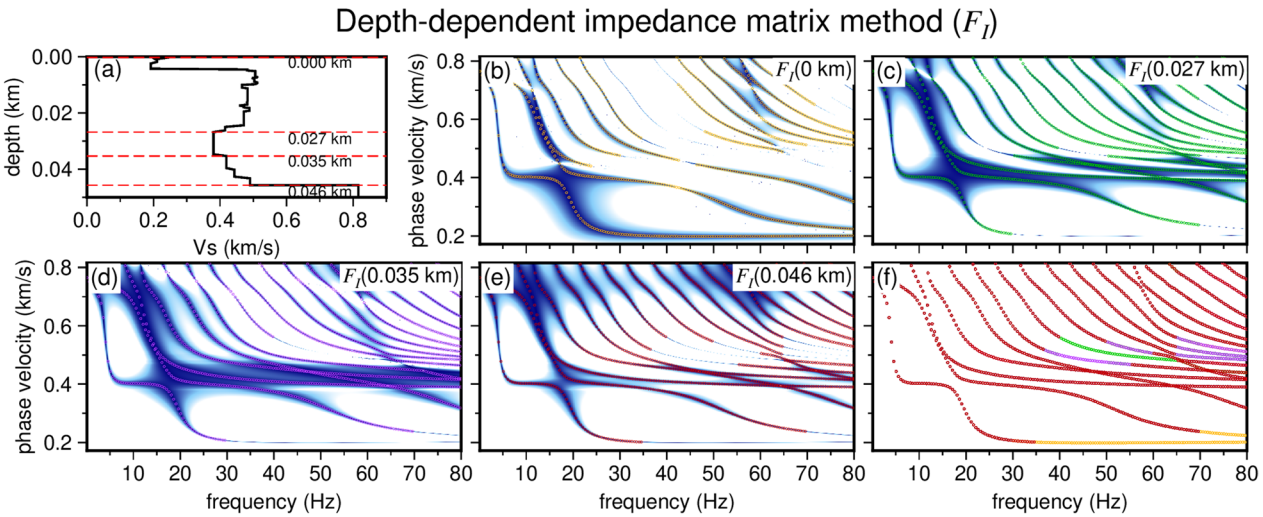
This study introduces a new impedance matrix method for calculating surface wave dispersion in stratified layered media, with particular attention to high-frequency scenarios of higher modes in the existence of a low-velocity layer where the missing root problem is most severe. The key innovation of the method is the introduction of depth-dependent dispersion equations, specifically designed to address and mitigate the issue of missing roots for the trapped modes. This method provides flexibility in the dispersion equation, enabling easier root detection and improving the resolution of certain roots. Extensive comparative analysis with other methods, including the compound matrix method and the generalized reflection and transmission coefficients method, highlights the practicality of our approach.

Keywords Theoretical seismology, Multimodal surface wave, High-frequency surface wave, Trapped mode, impedance matrix

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Graphical Abstract



1 Introduction

Seismic surface wave analysis is widely embraced for constructing near-surface shear-wave velocity models (Socco et al. 2010). The deployment of dense nodal arrays in recent years (Lin et al. 2013) and advancements in array-based techniques, such as the multichannel analysis of surface waves (MASW) (Park et al. 1999; Xia et al. 1999), and the frequency-Bessel transformation method (Forbriger 2003; Wang et al. 2019; Hu et al. 2020; Xi et al. 2021; Zhou and Chen 2021; Luo et al. 2022; Jiang et al. 2025), have led to the increasing availability of observations of surface wave dispersion curves of higher modes with higher frequencies. Additionally, the use of continuous data from distributed acoustic sensing has expanded the scope of high-frequency, higher-mode surface wave analysis (Zeng et al. 2017; Cheng et al. 2023). Furthermore, high-frequency higher-mode dispersions are essential for computing synthetic seismograms using the mode-summation method (Harkrider 1964).

The commonly adopted approach for computing theoretical dispersion curves involves two steps. First, the propagator matrix technique (Haskell 1953; Thomson 1950), or its modified version (Dunkin 1965; Thrower 1965; Schwab and Knopoff 1970; Kennett and Kerry 1979; Luco and Apsel 1983; Chen 1993; Liu 2010), is employed to calculate the dispersion function $F(f, c)$, where f and c represent frequency and phase velocity, respectively. Second, the roots of the dispersion equations $F(f, c) = 0$ are determined at each frequency f to construct the dispersion curve. For the first step, approaches such as the compound matrix approach (Dunkin 1965; Watson 1970; Wang and Herrmann 1980) and the reflection and transmission matrix approach (Kennett and Kerry 1979; Luco and Apsel 1983; Chen 1993) have been proposed to enhance the accuracy of dispersion equation calculations and mitigate precision loss at high frequencies. Buchen and Ben-Hador (1996) analyzed several such methods, concluding that they effectively address the high-frequency issue. In the second step, multiple methods are available to identify the roots of the dispersion equations. For example, the dichotomy method, which requires local monotonicity near the roots of the dispersion equation, is a proficient root-finding approach (Herrmann 2013). Newton's method has also been employed to locate the roots of dispersion equations (Anturkar et al. 1992). In other studies (Wu and Chen 2022; Zhang et al. 2024), the minima of the absolute value of the dispersion functions $|F(f, \omega)|$ were identified, and parabolic interpolation was applied to determine the corresponding roots. In the meantime, the asymptotic distribution of higher modes of elastic surface waves has been established (Fan et al. 2007; Jiang et al. 2024) to refine root-searching techniques. However, mode loss in models with low-velocity layers remains a challenge (He et al. 2006; Wu and Chen 2016), attributed to the trapped modes within these low-velocity zones (Wu and Chen 2016). To counter this, depth-dependent dispersion equations using the generalized reflection and transmission coefficients method (GRTM) (Chen 1993) have been developed (He et al. 2006; Wu and Chen 2016; Pan et al. 2022), exploiting the principle that dispersion

equations at different depths are sensitive to different modes, enabling detection of trapped modes at specific depths. This principle's applicability to other modified propagator matrix approaches warrants investigation.

The impedance matrix, defined as the ratio between the displacement and the stress, has been utilized in 1-D electromagnetic problems (Titchener and Wills 1991) and for calculating surface displacements in 1-D (Hager and Rostamian 1988; Caviglia and Morro 2000) and 2-D elastic media (Haines et al. 2004). This study further explores the impedance matrix's utility in computing the surface wave dispersion equation, deriving its explicit recursive relation between adjacent layers and the depth-dependent dispersion equation through the determinant's multiplicative property. A thorough comparison of the impedance matrix method with the compound matrix method and GRTM is conducted, suggesting its potential as an alternative for calculating higher-mode dispersion curves in horizontally stratified media at high frequencies.

2 Theory

2.1 Riccati equation of the impedance matrix

We start from the first-order wave equation of motion in a horizontally multi-layered medium (Zhu and Rivera 2002; Hu and Zhu 2017, 2018):

$$\frac{\mathrm{d}}{\mathrm{d}z}\begin{pmatrix}\mathbf{u}\\ \boldsymbol{\tau}\end{pmatrix}=\begin{pmatrix}\mathbf{M}_{11} & \mathbf{M}_{12}\\ \mathbf{M}_{21} & \mathbf{M}_{22}\end{pmatrix}\begin{pmatrix}\mathbf{u}\\ \boldsymbol{\tau}\end{pmatrix}, \tag{1}$$

where for the P-SV case, $\mathbf{u}=(U_r, U_z)^{\mathrm{T}}$ is the displacement, $\boldsymbol{\tau}=(T_z, T_r)^{\mathrm{T}}$ is the stress, and $\mathbf{M}_{ij}$ is the 2×2 submatrix of $\mathbf{M}$ matrix; for the SH case, $\mathbf{u}=U_t$, $\boldsymbol{\tau}=T_t$, and $\mathbf{M}_{ij}$ becomes scalar. The matrix $\mathbf{M}$ is depth-dependent and explicit expression of $\mathbf{M}$ can be found in Zhu and Rivera (2002) and in Appendix A.

The impedance matrix $\mathbf{Z}$, which draws a connection between the displacement and the stress, is defined as $\boldsymbol{\tau}=\mathbf{Z}\mathbf{u}$. Substituting this relation in the first-order wave equation of motion (Eq. 1), the impedance operator satisfies the Riccati equation:

$$\frac{\mathrm{d}\mathbf{Z}}{\mathrm{d}z}=\mathbf{Z}\mathbf{M}_{12}\mathbf{Z}+(\mathbf{M}_{22}\mathbf{Z}-\mathbf{Z}\mathbf{M}_{11})+\mathbf{M}_{21}. \tag{2}$$

Note that the definition of the impedance matrix in this study is slightly different from Haines et al. (2004). For the SH case, the corresponding Riccati equation is expressed as $\frac{\mathrm{d}Z}{\mathrm{d}z}=-\frac{1}{\mu}Z^2+\mu k^2-\rho\omega^2$ (see Appendix A for details). Here both μ and ρ are depth-dependent parameters.

The following boundary conditions of the impedance matrix $\mathbf{Z}$ need to be satisfied:

1. At the free surface, because $\boldsymbol{\tau}=0$, the determinant of the impedance matrix needs to be 0, that is

$$\det[\mathbf{Z}^{(0)}]=0. \tag{3}$$

2. In the halfspace, because there are no up-going waves, there is

$$\begin{pmatrix}\mathbf{u}^{(N+1)}\\ \boldsymbol{\tau}^{(N+1)}\end{pmatrix}=\begin{pmatrix}\mathbf{E}_{11}^{(N+1)}\Lambda_d(z_N)\mathbf{w}_d^{(N+1)}\\ \mathbf{E}_{21}^{(N+1)}\Lambda_d(z_N)\mathbf{w}_d^{(N+1)}\end{pmatrix}, \tag{4}$$

where $\mathbf{E}_{ij}^{(N+1)}$ is the submatrix of $\mathbf{E}$ for the halfspace, $\mathbf{w}$ is a constant vector which is determined by the boundary condition. $\mathbf{E}^{(N+1)}$ is determined by the model parameters of the halfspace, while Λ consists of elements associated with upward- and downward-propagating waves. These matrices are defined in Zhu and Rivera (2002), and the explicit expression of $\mathbf{E}$ and Λ can be found in Appendix A. Therefore,

$$\mathbf{Z}_N=\mathbf{E}_{21}^{(N+1)}\left(\mathbf{E}_{11}^{(N+1)}\right)^{-1}. \tag{5}$$

3. At the boundary between two layers, because $\mathbf{u}$ and $\boldsymbol{\tau}$ are continuous, the impedance matrix $\mathbf{Z}$ is also continuous.

2.2 Recursive relation of the impedance matrix

In a model with N layers over the halfspace (the halfspace is denoted by the $(N+1)$th layer). The Haskell propagator matrix in cylindrical, Cartesian, and spherical coordinates have been given in Zhu and Rivera (2002), Hu and Zhu (2017), Hu and Zhu (2024), and Bhattacharya (2025), respectively. The Haskell propagator matrix at the ith layer builds connections between the displacement-stress vectors for the $(i-1)$th layer and ith layer:

$$\begin{pmatrix}\mathbf{u}_{i-1}\\ \boldsymbol{\tau}_{i-1}\end{pmatrix}=\begin{pmatrix}\mathbf{a}_{11}^{(i)} & \mathbf{a}_{12}^{(i)}\\ \mathbf{a}_{21}^{(i)} & \mathbf{a}_{22}^{(i)}\end{pmatrix}\begin{pmatrix}\mathbf{u}_i\\ \boldsymbol{\tau}_i\end{pmatrix}, \tag{6}$$

where the $\mathbf{a}_{mn}^{(i)}$ is the submatrix of the propagator matrix defined in Zhu and Rivera (2002), Hu and Zhu (2017), or Hu and Zhu (2024) in different coordinates. The explicit expression of the propagator matrix $\mathbf{a}$ in the cylindrical coordinates is given in Appendix A. By substituting the expression of the impedance matrix $\boldsymbol{\tau}_i=\mathbf{Z}_i\mathbf{u}_i$ into Eq. (6), we have:

$$\mathbf{Z}_{i-1}=\left(\mathbf{a}_{21}^{(i)}+\mathbf{a}_{22}^{(i)}\mathbf{Z}_i\right)\left(\mathbf{a}_{11}^{(i)}+\mathbf{a}_{12}^{(i)}\mathbf{Z}_i\right)^{-1}. \tag{7}$$

The impedance matrix $\mathbf{Z}_N$ is directly computed at the halfspace (Eq. 5), enabling recursive calculation of the

impedance matrix for each layer from the bottom to the top of the model.

2.3 Depth-dependent dispersion equation

Equation (3) is the dispersion equation corresponding to the impedance matrix at the surface. Using the recursive relation in Eq. (7), combined with the multiplicative property of determinants $\det[\mathbf{XY}] = \det[\mathbf{X}]\det[\mathbf{Y}]$, we have:

$$\det[\mathbf{Z}_0] = \det\left[\mathbf{a}_{21}^{(1)} + \mathbf{a}_{22}^{(1)}\mathbf{Z}_1\right] / \det\left[\mathbf{a}_{11}^{(1)} + \mathbf{a}_{12}^{(1)}\mathbf{Z}_1\right]. \tag{8}$$

Therefore, the dispersion equation can be equally expressed as:

$$\det\left[\mathbf{a}_{21}^{(1)} + \mathbf{a}_{22}^{(1)}\mathbf{Z}_1\right] = 0. \tag{9}$$

Equation (9) is the dispersion equation corresponding to $\mathbf{Z}_1$.

Similarly to Eq. (9), we assume the dispersion equation involves impedance matrix corresponding to the impedance of the ith layer has the form:

$$\det[\mathbf{A}_i + \mathbf{B}_i\mathbf{Z}_i] = 0, \tag{10}$$

where $\mathbf{A}_i$ and $\mathbf{B}_i$ are matrices to be determined. Using Eq. (7) again, the following recursive relation can be derived (details of the derivation can be found in Appendix B):

$$\mathbf{A}_i = \mathbf{A}_{i-1}\mathbf{a}_{11}^{(i)} + \mathbf{B}_{i-1}\mathbf{a}_{21}^{(i)}, \tag{11}$$

$$\mathbf{B}_i = \mathbf{A}_{i-1}\mathbf{a}_{12}^{(i)} + \mathbf{B}_{i-1}\mathbf{a}_{22}^{(i)}, \tag{12}$$

and $\mathbf{A}_0 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, $\mathbf{B}_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Once $\mathbf{A}_i$ and $\mathbf{B}_i$ are computed, Eq. (10) provides a family of dispersion equations corresponding to the impedance matrix at each layer of the model. Because the velocity interfaces within the model are located at varying depths, the impedance matrix dispersion equations described by Eq. (10) for $i = 0, ..., N$ are depth-dependent.

In a matrix format, Eq. (11) and Eq. (12) can be written as:

$$\begin{pmatrix} \mathbf{A}_i & \mathbf{B}_i \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{i-1} & \mathbf{B}_{i-1} \end{pmatrix} \begin{pmatrix} \mathbf{a}_{11}^{(i)} & \mathbf{a}_{12}^{(i)} \\ \mathbf{a}_{21}^{(i)} & \mathbf{a}_{22}^{(i)} \end{pmatrix}. \tag{13}$$

Therefore,

$$\begin{pmatrix} \mathbf{A}_i & \mathbf{B}_i \end{pmatrix} = \begin{pmatrix} \mathbf{A}_0 & \mathbf{B}_0 \end{pmatrix} \mathbf{a}^{(1)} ... \mathbf{a}^{(i)}. \tag{14}$$

Defining the dispersion function

$$F_I^{(i)} = \det\left[\begin{pmatrix} \mathbf{A}_0 & \mathbf{B}_0 \end{pmatrix} \mathbf{a}^{(1)} ... \mathbf{a}^{(i)} \begin{pmatrix} \mathbf{I} \\ \mathbf{Z}_i \end{pmatrix}\right], \tag{15}$$

the dispersion equations can be written as $F_I^{(i)} = 0$, where i ranges from 0 to N.

3 Implementation

The propagator matrix $\mathbf{a}^{(i)}$ and the impedance matrix $\mathbf{Z}_i$ are functions of the angular frequency ω and the wavenumber k (see Appendix A for details). For a particular frequency f and phase velocity c, there are relations $f = \frac{\omega}{2\pi}$, $c = \frac{\omega}{k}$. The impedance matrix at the free surface can be calculated using the recursive relation in Eq. (7), starting from $\mathbf{Z}_N$ (Eq. (5)). The matrix associated with the impedance matrix dispersion function defined in Eq. (15) can be split into two parts: $\left[\begin{pmatrix} \mathbf{A}_0 & \mathbf{B}_0 \end{pmatrix} \mathbf{a}^{(1)} ... \mathbf{a}^{(i)}\right]$ and $\begin{pmatrix} \mathbf{I} \\ \mathbf{Z}_i \end{pmatrix}$. We compute these two parts in separate loops: the first loop iterates from the top to the bottom of the model, while the second loop iterates from the bottom to the top. By storing intermediate results, the dispersion functions $F_I^{(i)}$ can be efficiently evaluated at all depths (Fig. 1). Note that a virtual layer can be inserted into the model at any arbitrary depth z. Consequently, the dispersion equation can also be expressed as $F_I(z; f, c) = 0$, as it can be evaluated at any given depth z.

One computational issue is the loss of precision at high frequencies attributed to the multiplications of propagator matrices in Eq. (15). To address this, we utilize the compound matrix method to enhance stability in solving the PSV problem. Specifically, we adopt the algorithm of Wang and Herrmann (1980) to stabilize the computation. Numerical tests indicate that the SH problem is free from stability issues (see our numerical examples in the supplementary material).

To further stabilize the computation, the exponential growing terms can be factored out in the propagator matrix. For instance, for the *SH* case, using the explicit expression of the matrix $\mathbf{E}$, the recursive relation in Eq. (7) is simplified as $Z_n = \frac{-\mu X_\beta + C_\beta Z_{n+1}}{C_\beta - (Y_\beta/\mu) Z_{n+1}}$, where $C_\beta = \cosh(\nu_\beta d)$, $X_\beta = \nu_\beta \sinh(\nu_\beta d)/k$, $Y_b = k \sinh(\nu_\beta d)/\nu_\beta$, and $\nu_\beta = \sqrt{k^2 - (\omega/\beta)^2}$. An extra term $e^{-\mathrm{Re}(\nu_\beta)\mathrm{d}}$ can be multiplied both at the numerator and denominator, and the high-frequency problem can be mitigated.

4 Numerical experiments

In the main text, we focus on numerical experiments on Rayleigh wave dispersion equations for several models. Corresponding numerical experiments for Love wave dispersion equations are provided in the supplementary material (Figs. S1-S6).

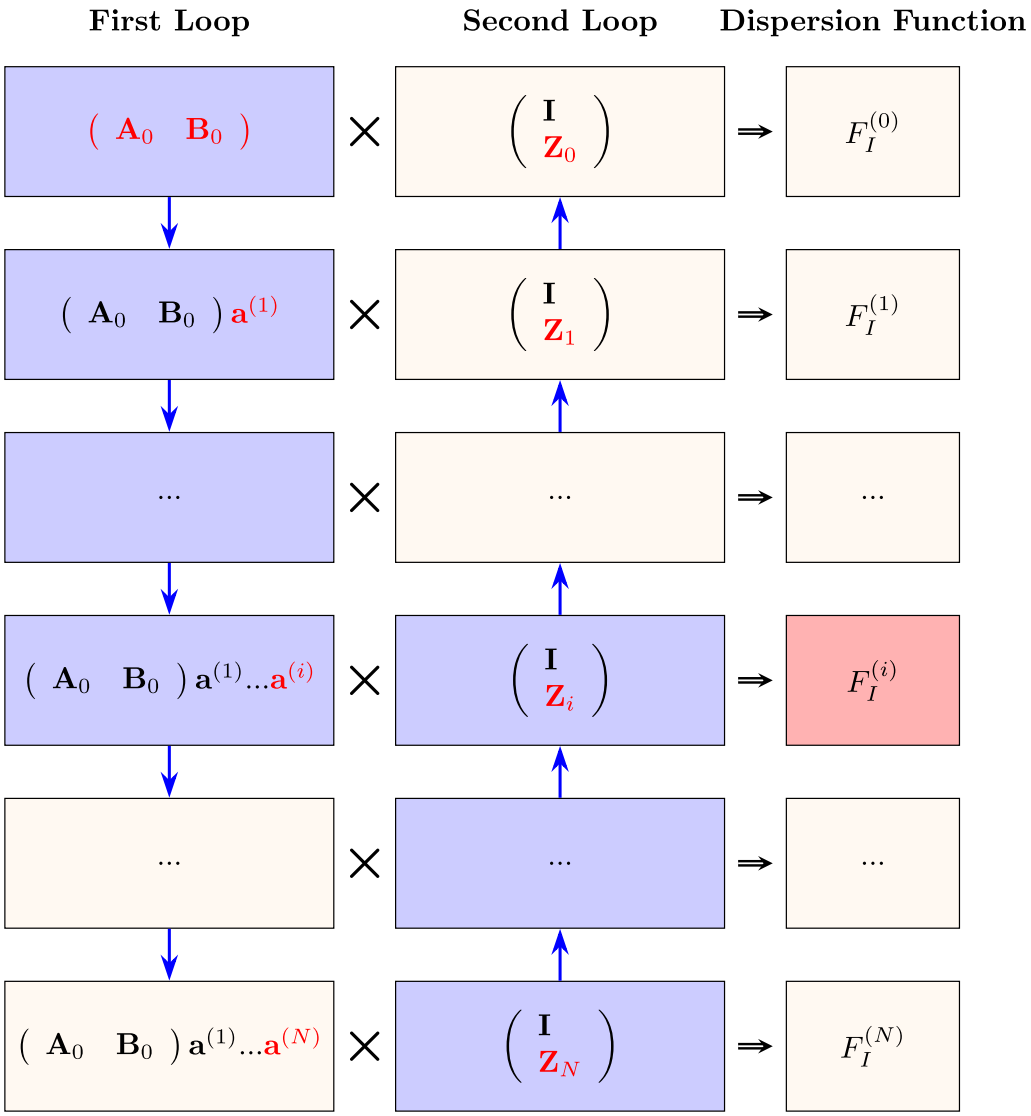


Fig. 1 Workflow for calculating the depth-dependent impedance matrix dispersion equations. The dispersion function at the *i*th layer is described by Eq. (15). The matrix associated with the determinant can be split into two parts: $(\mathbf{A}_0 \quad \mathbf{B}_0)\mathbf{a}^{(1)}...\mathbf{a}^{(i)}$ and $(\mathbf{I} \quad \mathbf{Z}_i)$. These two parts are computed in separate loops: the first loop iterates from the top to the bottom of the model, while the second loop iterates from the bottom to the top. By storing intermediate results, the dispersion functions $F_I^{(i)}$ can be efficiently evaluated at all depths. Symbols in red indicate the newly computed matrix at each depth

4.1 A multilayered crustal model

We evaluated the impedance matrix using a multilayered crustal model shown in Table 1, which features a crust with a thickness of 30 km (Chen 1993). Initially, we calculated the dispersion functions ($F_I^{(0)}(f,c)$) for the Rayleigh wave (Fig. 2a) at f =1 Hz using the impedance matrix method. We then compared these results with those obtained using the compound matrix method ($F_C(f,c)$) (Wang and Herrmann 1980) (Fig 2b) and the original GRTM ($F_G(f,c)$)) (Chen 1993) (Fig. 2c). The dispersion functions were calculated and plotted over a phase velocity range of $[0.9\beta_{\min}, 1.0\beta_{\max}]$ with an increment of $\Delta c = 10^{-3}$ km/s, where $\beta_{\min}$ and $\beta_{\max}$ represent the

Table 1 Velocity parameters of Model 1

Thickness (km)	β (km/s)	α (km/s)	ρ (g/cm^3)
18.0	3.50	6.00	2.80
6.0	3.65	6.30	2.90
6.0	3.90	6.70	3.10
∞	4.70	8.20	3.30

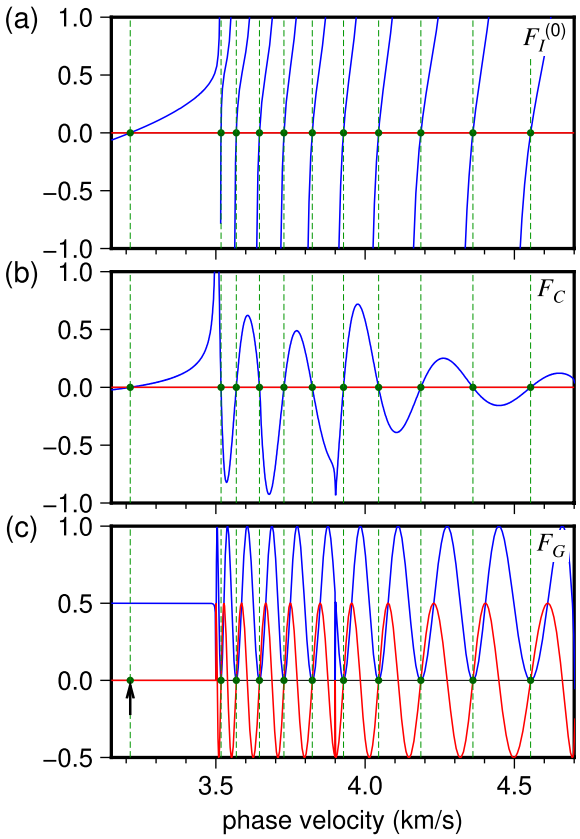


Fig. 2 Dispersion functions of Rayleigh waves for Model 1 at the frequency of 1 Hz, computed through **a** the impedance matrix method ($F_I^{(0)}$), **b** the compound matrix method (F_C), and **c** the GRTM (F_G). The blue lines and red lines are the real and imaginary parts of the dispersion functions, respectively. The locations of the modes are denoted as green dots. The black arrow highlights a potential omission of the fundamental mode in the original GRTM method

minimum and maximum shear-wave velocities of the model, respectively. The concurrence of root locations of the dispersion equation (solid green dots in Fig. 2) across these methods confirms the accuracy of the impedance matrix method. However, it is noteworthy that, as discussed in Chen (1993), the fundamental mode of Rayleigh waves is frequently overlooked in the original GRTM, necessitating a modified version for surface wave analysis (He et al. 2006; Wu and Chen 2016).

Subsequently, we calculated depth-dependent dispersion functions using the impedance matrix method ($F_I^i(f,c)$), for $i = 0, 1, 2, 3$. Fig 3 presents the dispersion functions for Rayleigh waves, demonstrating consistent roots across all dispersion equations, thereby affirming the reliability of the impedance matrix method for depth-dependent analyses.

4.2 Multilayered model with a low-velocity layer

The near surface in a sedimentary area can be usually described as a model with a low-velocity layer. For the second test, we consider a multilayered model with a low-velocity layer (Model 2, Table 2), as explored by Shi et al. (2022). Similar to the first test, we compared the dispersion function calculated using the impedance matrix ($F_I^{(0)}(f,c)$) at f =20 Hz for the Rayleigh waves (Fig. 4a) with those obtained via the compound matrix method ($F_C(f,c)$) (Fig. 4b) and the GRTM ($F_G(f,c)$) (Fig. 4c). Additionally, we also calculated the depth-dependent dispersion functions for the Rayleigh waves using the impedance matrix (Fig. 5).

This test demonstrated that both the impedance matrix method and the GRTM tend to miss the second mode (as indicated by arrows in Fig. 4a, c). However, employing depth-dependent dispersion functions can mitigate this oversight, since dispersion equations at varying depths can provide compensatory data (Fig. 5).

5 Discussion

The results displayed in Fig. 5 illustrate that depth-dependent dispersion equations effectively address mode omission issues. We delve deeper into the causes of mode loss and the efficacy of depth-dependent dispersion equations in resolving these issues. Theoretically, all dispersion equations should encompass a complete set of eigenvalues (Wu and Chen 2016). However, practical limitations, such as the different specific forms of dispersion equations and the finite searching step Δc, can make some eigenvalues challenging to evaluate. An extreme case is the fundamental Rayleigh mode discussed in Chen (1993), near which the dispersion function $F_G(f,c)$ varies dramatically, hindering direct numerical computation of the root (see Fig. 3c). Conversely, roots corresponding to gently varying dispersion functions are more readily identified, especially when the derivative-based method (e.g., the Newton–Raphson method (Anturkar et al. 1992; Hooshyar 2025)) is used (Fig. S7). This is because the mild variations of dispersion function near the roots make it more robust to the choice of initial guesses, facilitating easier root detection. To show it clearly, we calculated and plotted the Rayleigh wave dispersion functions using the impedance matrix $F_I^{(i)}(f,c)$ across the frequency range of 1–60 Hz for Model 2 (Fig. 6). In these frequency–phase spectrograms, modes associated with mild variations in the dispersion

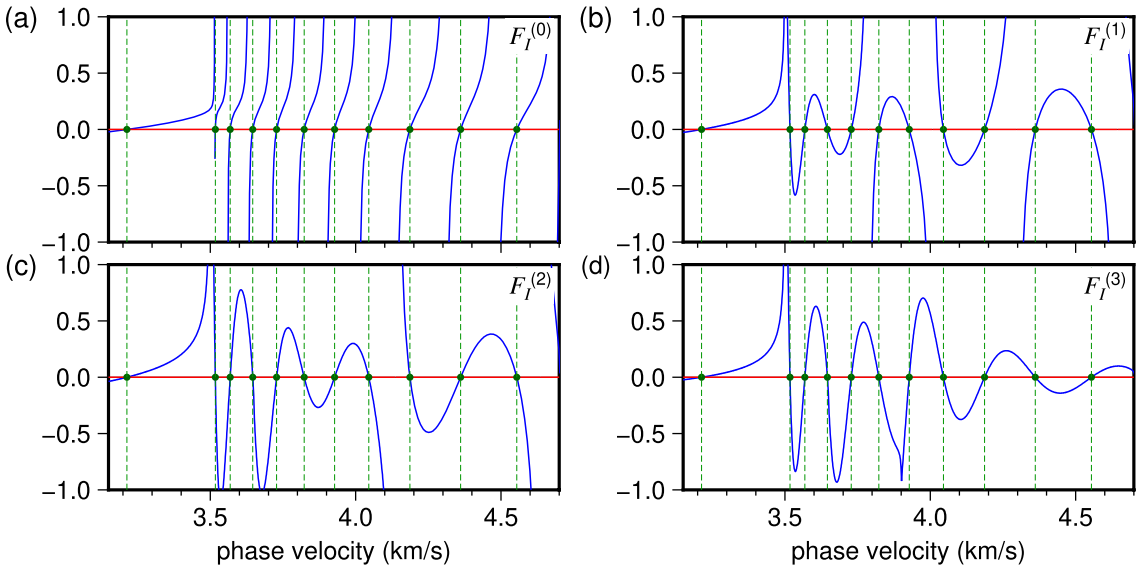


Fig. 3 Depth-dependent Rayleigh wave dispersion functions for Model 1 at the frequency of 1 Hz, derived via the impedance matrix method ($F_I^{(i)}$) at the *i*th layer (i =0,1,2,3). **a**–**d** Illustrate the matrix dispersion functions at the free surface, the 1st interface, the 2nd interface, the 3rd interface, respectively. The blue lines and red lines represent the real and imaginary parts of the dispersion functions, respectively. The locations of the modes are indicated by green dots

Table 2 Velocity parameters of Model 2

Thickness (km)	β (km/s)	α (km/s)	ρ (g/cm^3)
0.015	0.3	0.8	1.8
0.015	0.4	1.0	1.9
0.015	0.3	0.8	1.8
∞	0.5	1.2	2.0

function are easily discernible (highlighted by dashed red rectangles in Fig. 6). For instance, $F_I^{(0)}$ shows gentle variations around the fundamental Rayleigh mode (Fig. 6a), and $F_I^{(3)}$ exhibits similar behavior at higher modes (Fig. 6d). The results indicate that the mild variations around the roots generally correspond to a wider range of viable initial guesses for a certain root (Fig. S8). Thus, depth-dependent spectrograms collaboratively provide a comprehensive set of modes.

Furthermore, we calculated the depth-dependent dispersion functions at f =20 Hz using both the impedance matrix method and modified GRTM (He et al. 2006; Wu and Chen 2016) for Rayleigh wave (Fig. 7). This comparison between the impedance matrix method and the modified GRTM confirms that both approaches effectively yield a complete set of modes. Notably, the impedance matrix method exhibits two advantages: first, the fundamental mode, particularly for Rayleigh waves, is more readily identifiable (Fig. 7a); second, critical modes (Wang and Lu 2023), which correlate with the shear-wave velocity of certain model layers (marked by solid red

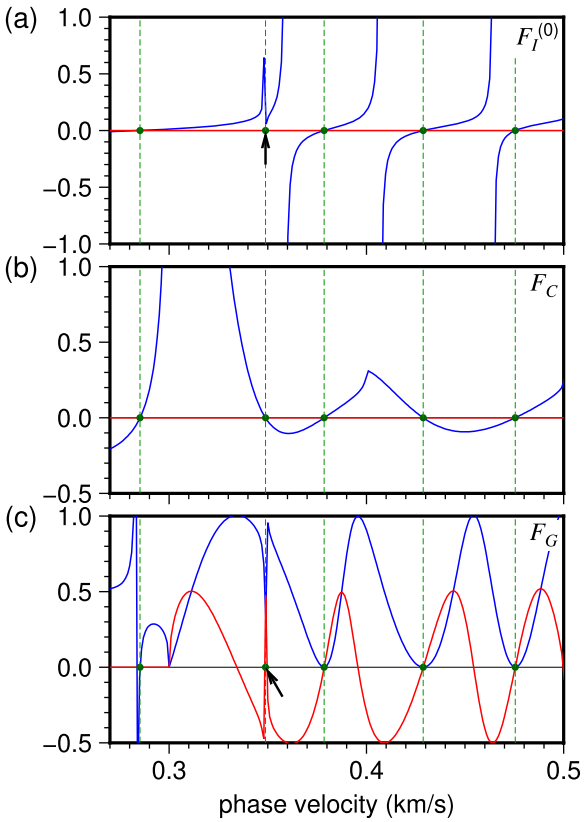


Fig. 4 Rayleigh wave dispersion functions for Model 2 at the frequency of 20 Hz, obtained using a the impedance matrix method ($F_I^{(0)}$), b the compound matrix method (F_C), and c the GRTM (F_G). The solid blue and red lines represent the real and imaginary parts, respectively, with solid green dots indicating the roots of dispersion equations. Black arrows indicate the mode to be probably missed in the impedance matrix method ($F_I^{(0)}$) and original GRTM (F_G)

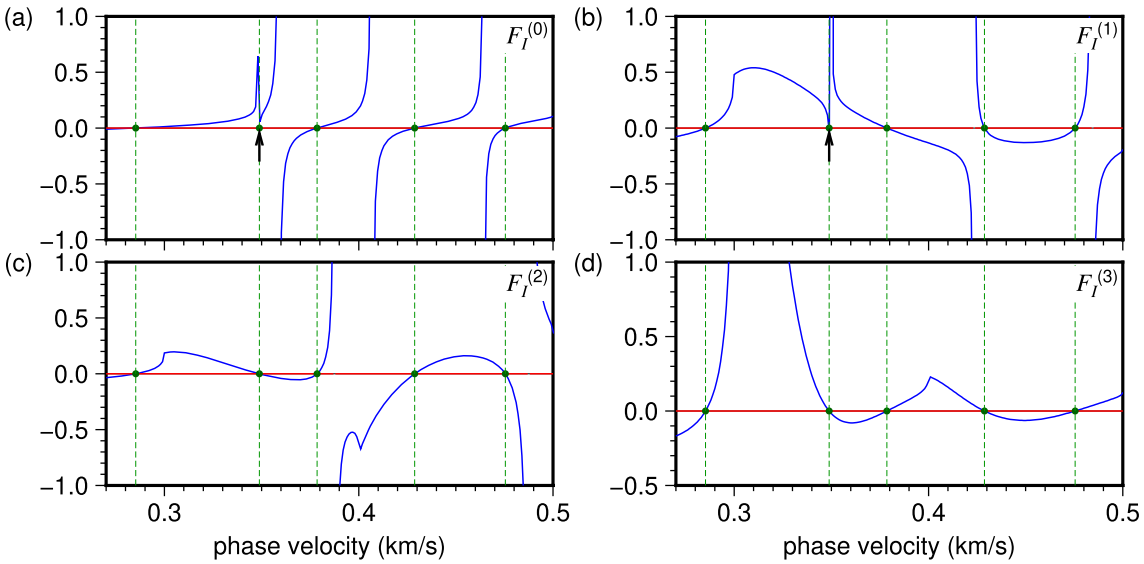


Fig. 5 Rayleigh wave depth-dependent dispersion functions for Model 2 at the frequency of 20 Hz, analyzed using the impedance matrix method ($F_I^{(i)}$) at the i—th layer (i =0,1,2,3). The blue lines and red lines are the real and imaginary parts of the dispersion functions, respectively. The locations of the modes are denoted as green dots. Black arrows indicate the mode to be probably missed using F_I at certain depths. The dispersion functions at all depths collaboratively provide a comprehensive set of modes

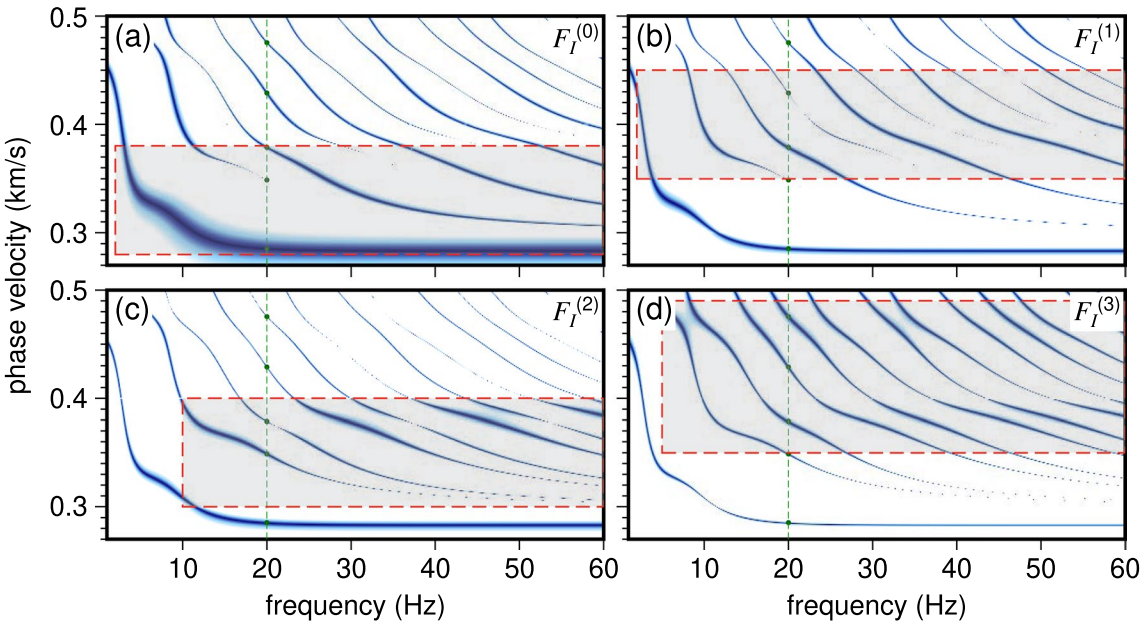


Fig. 6 Dispersion functions of Rayleigh waves for Model 2 across frequencies from 1-60 Hz (0.1 Hz intervals), calculated using the impedance matrix method ($F_I^{(i)}$) for ith layers (i =0,1,2,3). **a**–**d** Depict the dispersion functions at the free surface, the 1st interface, the 2nd interface, and the 3rd interface, respectively. The color indicates the norm value of the functions (the closer the value approaches 0, the darker the color), with a dashed red rectangle highlighting easily identifiable modes

rectangles in Fig. 7b), are absent in the impedance matrix method.

In a more complex model containing two low-velocity layers, as shown in Table 3, we calculated the dispersion functions of Rayleigh wave at 1.8 Hz using both the depth-dependent impedance matrix method (F_I) (Fig. 8a,b,c) and the compound matrix method (F_C) (Fig. 8d). It is shown the 10th and 11th modes are located very close to each other around 4.124 km/s, making it easy to miss one of them (see red arrows in

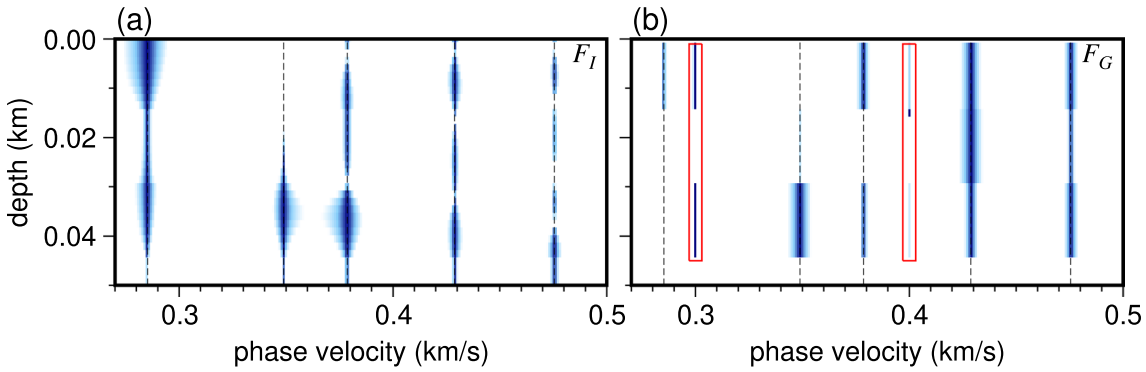


Fig. 7 Depth-dependent Rayleigh wave dispersion functions for Model 2 at 20 Hz, compared using a the impedance matrix method (F_I) and b the modified GRTM (F_G). Model 2 is discretized in 0.0015-km intervals in depth, with a solid red rectangle identifying critical modes correlating to shear-wave velocity in certain layers

Fig. 8a, d). In a magnified view of F_C around 4.124 km/s (Fig. 8e and f), it is evident the two modes are difficult to be separated because the values of F_C between these two modes are almost constant (approximately equal 0). In contrast, the zoomed-in view of F_I near the two roots (Fig. 8b) and the corresponding F_I at the depth of 35 km (Fig. 8c) show that the dispersion functions between the two roots have obvious lateral variation. Figure S9 demonstrates that the resolution of the roots improves with large variations in the dispersion function, particularly given the finite precision of numerical computation. This enhancement is especially critical in scenarios where two roots are in close proximity. The depth-dependent dispersion function F_I enhances the resolution of phase velocity, facilitating the identification of closely spaced roots. It is important to note that this argument applies to all root-searching methods, including the dichotomy method and the Newton–Raphson method.

Additionally, we test the depth-dependent impedance matrix dispersion equations in a realistic model in the Apennine orogenic belt. This model was constructed by Albarello et al. (2023) based on seismic microzonation studies conducted in Southern Italy by local authorities, under the coordination of the Italian Centre for Seismic Microzonation and Applications (Moscatelli et al. 2020). The model, shown in Fig. 9a, contains 109 layers. In this test, the impedance matrix dispersion equations at various depths were calculated (Fig. 9b–e). It is observed that the dispersion equations at different depths are sensitive to different portions of the dispersion spectrum (Fig. 9b–e). Together, these dispersion equations yield a complete set of surface wave dispersion curves (Fig. 9f).

We also compare the dispersion functions calculated using the compound matrix method and the GRTM method (see Fig. 10a, b). The dispersion functions obtained by the compound matrix method are not sensitive to certain modes (see the red dotted rectangle in Fig. 10a), whereas these modes are clearly identified by the depth-dependent impedance matrix dispersion equations (Fig. 9f). Compared to the GRTM, the depth-dependent impedance matrix dispersion functions are free from the critical modes (see the red dotted rectangles in Fig. 10b).

We note that the modified Thomson–Haskell matrix method proposed by Liu (2010) is mathematically equivalent to, but is in different form from, the impedance matrix dispersion function at the surface ($F_I^{(0)}$). From numerical tests, Liu (2010) demonstrated that the computational speed of the modified Thomson–Haskell matrix method increases by more than 15% compared to the GRTM with the speed-up algorithm of Pei et al. (2008). The computation of $F_I^{(0)}$ should be as fast as the modified Thomson–Haskell matrix method. In our implementation, the computational cost of impedance matrix dispersion functions at other depths, $F_I(z)$ $(z \neq 0)$, is nearly identical to that of F_I^0. As shown in Fig. 9, only the impedance matrix dispersion equations at a few depths are usually needed. Furthermore, leveraging the asymptotic eigenvalue distribution of the dispersion equations (Fan et al. 2007; Jiang et al. 2024), more

Table 3 Velocity parameters of Model 3

Thickness (km)	β (km/s)	α (km/s)	ρ (g/cm^3)
10	3.8	6.5	2.7
10	3.5	6.0	2.5
10	4.4	7.5	3.1
10	4.1	7.0	2.9
∞	4.7	8.0	3.3

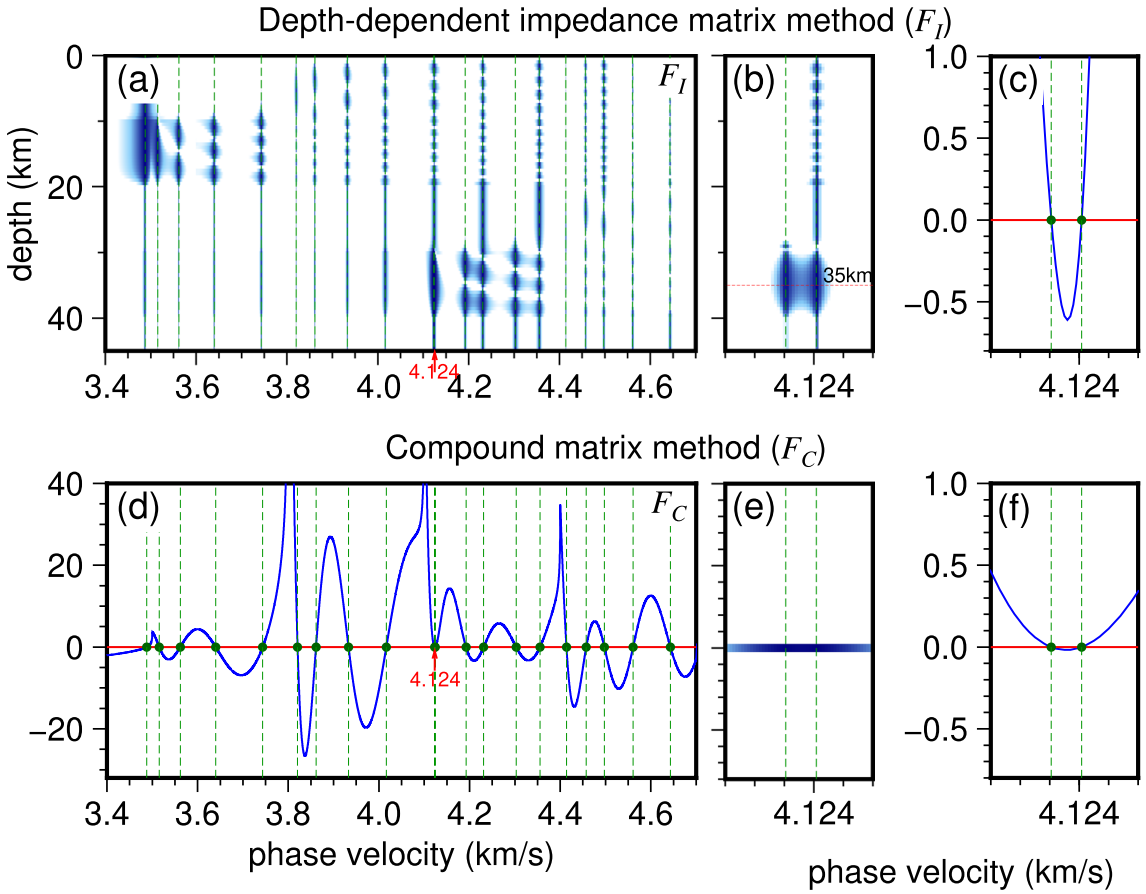


Fig. 8 Dispersion function of Rayleigh waves for Model 3 at 1.8 Hz using **a** the depth-dependent impedance matrix method (F_I), and **d** the compound matrix method (F_C). **b** Is the magnified view of F_I near the 10th and 11th roots (shown by red arrow in **a**), and **c** is the corresponding F_I at the depth of 35 km. **e** and **f** Are the magnified view of F_C near the 10th and 11th roots (shown by red arrow in **d**) with phase velocity between 4.121 km/s and 4.126 km/s

sophisticated strategies can be designed. In this case, only additional dispersion equations in certain frequency ranges are required, as implemented by Wu and Chen (2022) for dispersion equations using depth-dependent GRTM.

Selecting the optimal depth for dispersion function calculations is not a strictly deterministic process. However, based on our experience, we have established several practical guidelines: (1) depths at the top of the model and below low-velocity layers are particularly effective in addressing the missing root issue (Fig. 9); (2) the fundamental mode is generally most sensitive to the dispersion function at surface or near-surface depths ($F_I^{(0)}$) (Fig. 6).

We also observe that the compound matrix dispersion function F_C, as implemented by Herrmann (2013), and the impedance matrix dispersion function at halfspace $F_I^{(N)}$ share the same functional form. The only difference lies in a constant factor, which arises from the use of $\mathbf{Z}_N$ for boundary condition in our approach, whereas Herrmann (2013) employs a different expression. The similarity is expected, as both methods stabilize calculations by leveraging products of the compound matrix rather than directly utilizing the Haskell matrix. Consequently, the depth-dependent $F_I^{(i)}$ can be interpreted as and extension of the compound matrix method, offering enhanced flexibility to address the mode-missing problem, as demonstrated in Figs. 6, 8 and S8.

6 Conclusion

In this study, we employ the impedance matrix to calculate depth-dependent surface wave dispersion equations in stratified media. We have derived the explicit recursive relationship of the impedance matrix between adjacent layers. This method provides flexibility in the dispersion equation, allowing for easier root detection

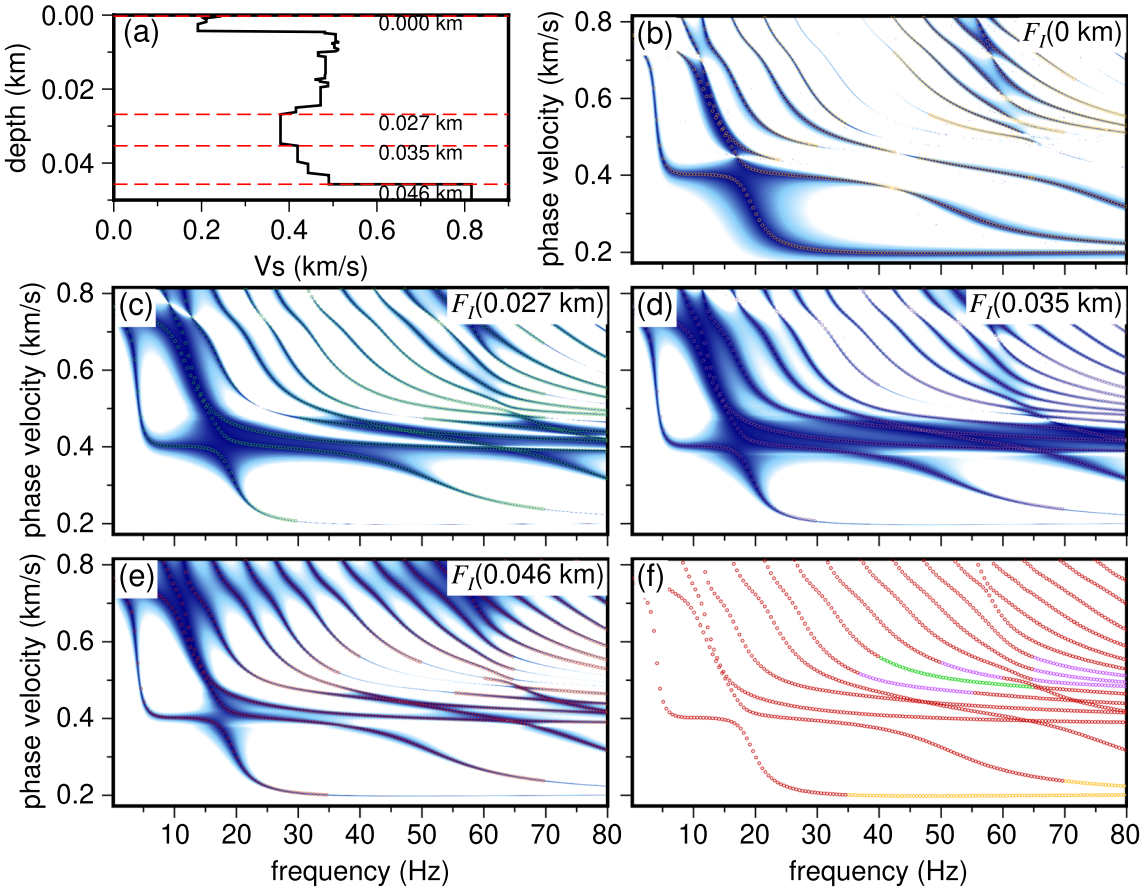


Fig. 9 Application of the depth-dependent impedance matrix method to the realistic model in southern Italy. **a** The realistic V_S model in southern Italy. The density ρ was set to be 1.8 g/cm^3, and V_P was calculated according to V_S using the empirical relations and provided by Albarello et al. (2023). **b**–**e** Are the dispersion functions of the model at the depths of 0 km, 0.027 km, 0.035 km, and 0.046 km, respectively. The circles in **b**–**e** in different colors are the modes that are easily detectable. The easily detectable modes at different depths altogether result in a complete set of modes in **f**

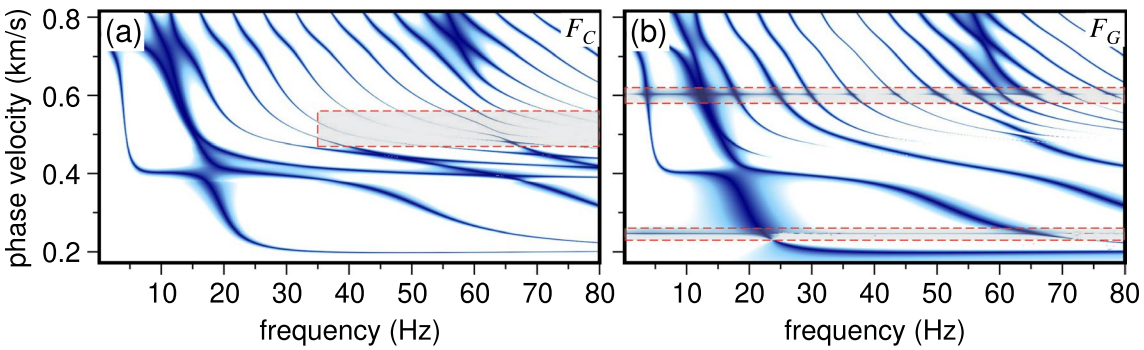


Fig. 10 Application of the compound matrix method and the original GRTM to the realistic model in southern Italy. **a** The dispersion function of the compound matrix method; **b** the dispersion function of the original GRTM. The region in dashed red line in a is where the modes are not easily detectable. The region in dashed red line in b is where there are critical modes

and improving the resolution of certain roots. Our tests reveal that a complete set of modes, which are often missed in the presence of a low-velocity layer, can be successfully obtained by integrating depth-dependent dispersion equations. Comparative analyses with other methods, such as the compound matrix method and the GRTM, demonstrate the effectiveness of the impedance matrix approach. The impedance matrix method is particularly adept at computing higher-mode surface wave dispersion curves at high frequencies.

Explicit formulations of related matrices in cylindrical coordinates

In this appendix, we provide the explicit formulations of **M**, **E**, Λ, and **a** in a horizontally layered medium in the cylindrical coordinates (Zhu and Rivera 2002). The formulations for the corresponding matrices in Cartesian and spherical coordinates can be found in Hu and Zhu (2017), Hu and Zhu (2024), and Bhattacharya (2025).

For the PSV wave:

$$\mathbf{M}=\begin{bmatrix} 0 & -1 & 0 & \frac{1}{\mu} \\ 1-2\xi & 0 & \frac{\xi}{\mu} & 0 \\ 0 & -\rho\left(\frac{\omega}{k}\right)^2 & 0 & 1 \\ 4\mu\,\xi_1-\rho\left(\frac{\omega}{k}\right)^2 & 0 & 2\xi-1 & 0 \end{bmatrix}, \tag{16}$$

$$\mathbf{E}=\begin{bmatrix} -1 & -\frac{\nu_\beta}{k} & 1 & \frac{\nu_\beta}{k} \\ \frac{\nu_\alpha}{k} & 1 & \frac{\nu_\alpha}{k} & 1 \\ -2\mu\gamma_1 & -2\mu\frac{\nu_\beta}{k} & 2\mu\gamma_1 & 2\mu\frac{\nu_\beta}{k} \\ 2\mu\frac{\nu_\alpha}{k} & 2\mu\gamma_1 & 2\mu\frac{\nu_\alpha}{k} & 2\mu\gamma_1 \end{bmatrix}, \tag{17}$$

$$\Lambda=\begin{bmatrix} e^{-\nu_\alpha z} & 0 & 0 & 0 \\ 0 & e^{-\nu_\beta z} & 0 & 0 \\ 0 & 0 & e^{\nu_\alpha z} & 0 \\ 0 & 0 & 0 & e^{\nu_\beta z} \end{bmatrix}, \tag{18}$$

and

$$\mathbf{a}=\gamma\begin{bmatrix} C_\alpha-\gamma_1 C_\beta & \gamma_1 Y_\alpha-X_\beta & \frac{C_\beta-C_\alpha}{2\mu} & \frac{X_\beta-Y_\alpha}{2\mu} \\ \gamma_1 Y_\beta-X_\alpha & C_\beta-\gamma_1 C_\alpha & \frac{X_\alpha-Y_\beta}{2\mu} & \frac{C_\alpha-C_\beta}{2\mu} \\ 2\mu\gamma_1\left(C_\alpha-C_\beta\right) & 2\mu\left({\gamma_1}^2 Y_\alpha-X_\beta\right) & C_\beta-\gamma_1 C_\alpha & X_\beta-\gamma_1 Y_\alpha \\ 2\mu\left({\gamma_1}^2 Y_\beta-X_\alpha\right) & 2\mu\gamma_1\left(C_\beta-C_\alpha\right) & X_\alpha-\gamma_1 Y_\beta & C_\alpha-\gamma_1 C_\beta \end{bmatrix}. \tag{19}$$

For the SH wave:

$$\mathbf{M}=\begin{bmatrix} 0 & \frac{1}{\mu} \\ \mu-\rho\left(\frac{\omega}{k}\right)^2 & 0 \end{bmatrix}, \tag{20}$$

$$\mathbf{E}=\begin{bmatrix} -1 & 1 \\ \mu\frac{\nu_\beta}{k} & \mu\frac{\nu_\beta}{k} \end{bmatrix}, \tag{21}$$

$$\Lambda=\begin{bmatrix} e^{-\nu_\beta z} & 0 \\ 0 & e^{\nu_\beta z} \end{bmatrix}, \tag{22}$$

and

$$\mathbf{a}=\begin{bmatrix} C_\beta & -\frac{Y_\beta}{\mu} \\ -\mu X_\beta & C_\beta \end{bmatrix}. \tag{23}$$

In the above equations in this appendix, α, β, and ρ are the compressional-wave velocity, shear-wave velocity, and density. $d=z_{n-1}-z_n$, $\xi=\frac{\beta^2}{\alpha^2}$, $\xi_1=1-\xi$, $\mu=\rho\beta^2$, $\nu_\alpha=\sqrt{k^2-\left(\frac{\omega}{\alpha}\right)^2}$, $\nu_\beta=\sqrt{k^2-\left(\frac{\omega}{\beta}\right)^2}$, $\gamma=2k^2\beta^2/\omega^2$, $\gamma_1=1-\frac{1}{\gamma}$, $C_\alpha=\cosh(\nu_\alpha d)$, $C_\beta=\cosh(\nu_\beta d)$, $X_\alpha=\nu_\alpha\sinh(\nu_\alpha d)/k$, $X_\beta=\nu_\beta\sinh(\nu_\beta d)/k$, $Y_\alpha=k\sinh(\nu_\alpha d)/\nu_\alpha$, and $Y_\beta=k\sinh(\nu_\beta d)/\nu_\beta$.

Using the explicit expression of **M** in Eq. (24), the Ricatti equation of SH wave in Eq. (2) can be derived as:

$$\frac{dZ}{dz}=-\frac{1}{\mu}Z^2+\mu k^2-\rho\omega^2. \tag{24}$$

Note that μ and ρ in Eq. (24) are depth-dependent parameters.

Derivation of matrix A_i and B_i in Eq. (10)

Similar to Eq. (9), we assume the dispersion equation involves impedance matrix of the $(i-1)$th layer has the form:

$$\det[\mathbf{A}_{i-1}+\mathbf{B}_{i-1}\mathbf{Z}_{i-1}]=0. \tag{25}$$

Therefore, the dispersion equation involves the ith layer impedance matrix has the form:

$$\det[\mathbf{A}_i+\mathbf{B}_i\mathbf{Z}_i]=0. \tag{26}$$

Substituting the recursive relation

$$\mathbf{Z}_{i-1} = \left(\mathbf{a}_{21}^{(i)} + \mathbf{a}_{22}^{(i)}\mathbf{Z}_i\right)\left(\mathbf{a}_{11}^{(i)} + \mathbf{a}_{12}^{(i)}\mathbf{Z}_i\right)^{-1} \tag{B12}$$

into Eq. (25), we have

$$\det\left[\left(\mathbf{A}_{i-1}\mathbf{a}_{11}^{(i)} + \mathbf{B}_{i-1}\mathbf{a}_{21}^{(i)}\right) + \left(\mathbf{A}_{i-1}\mathbf{a}_{12}^{(i)} + \mathbf{B}_{i-1}\mathbf{a}_{22}^{(i)}\right)\mathbf{Z}_i\right] / \det\left[\mathbf{a}_{11}^{(i)} + \mathbf{a}_{12}^{(i)}\mathbf{Z}_i\right] = 0. \tag{28}$$

Therefore, the dispersion equation for the ith layer is

$$\det\left[\left(\mathbf{A}_{i-1}\mathbf{a}_{11}^{(i)} + \mathbf{B}_{i-1}\mathbf{a}_{21}^{(i)}\right) + \left(\mathbf{A}_{i-1}\mathbf{a}_{12}^{(i)} + \mathbf{B}_{i-1}\mathbf{a}_{22}^{(i)}\right)\mathbf{Z}_i\right] = 0. \tag{29}$$

Comparing Eqs. (26) and (29), we have:

$$\mathbf{A}_i = \mathbf{A}_{i-1}\mathbf{a}_{11}^{(i)} + \mathbf{B}_{i-1}\mathbf{a}_{21}^{(i)}, \tag{30}$$

$$\mathbf{B}_i = \mathbf{A}_{i-1}\mathbf{a}_{12}^{(i)} + \mathbf{B}_{i-1}\mathbf{a}_{22}^{(i)}. \tag{31}$$

Choosing $i = 1$, Eqs. (30) and (31) become

$$\mathbf{A}_1 = \mathbf{A}_0\mathbf{a}_{11}^{(1)} + \mathbf{B}_0\mathbf{a}_{21}^{(1)}, \tag{32}$$

$$\mathbf{B}_1 = \mathbf{A}_0\mathbf{a}_{12}^{(1)} + \mathbf{B}_0\mathbf{a}_{22}^{(1)}. \tag{33}$$

From Eq. (9), $\mathbf{A}_1 = \mathbf{a}_{21}^{(1)}$, and $\mathbf{B}_1 = \mathbf{a}_{22}^{(1)}$. Therefore, from Eqs. (32) and (33), we have $\mathbf{A}_0 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{B}_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

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Author contributions
XJ and SH derived the formulation, wrote the corresponding computer code, and drafted the manuscript. XJ performed the numerical tests. SL, RH, and HX analyzed the numerical results and revised the manuscript. All authors read and approved the final manuscript.

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Declarations

Competing interests
The authors acknowledge there are no competing interests recorded.

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